a special robot loaded with a customised channel sounder and other equipment.

NIST study results confirm that narrow beams can significantly reduce signal interference and delays, and that an optimised beam orientation reduces energy loss during transmissions.

Shape-shifting origami could help antenna systems adapt on-the-fly

Researchers at Georgia Institute of Technology, USA, have devised a method for using an origami-based structure to create radio frequency (RF) filters that have adjustable dimensions, enabling devices to change signals these devices block throughout a large range of frequencies.

The new approach for creating these tunable filters could have a variety of uses, from antenna systems capable of adapting in real time to ambient conditions, to next-generation electromagnetic cloaking systems that could be reconfigured on-the-fly to reflect or absorb different frequencies.

The team focussed on a particular pattern of origami, called Miura-Ori, which has the ability to expand and contract like an accordion. Glaucio Paulino, the Raymond Allen Jones Chair of Engineering and professor in Georgia Tech School of Civil and Environmental Engineering, says, "Miura-Ori pattern has an infinite number of possible positions along its range of extension from fully compressed to fully expanded. A spatial filter made in this fashion can achieve similar versatility, changing which frequency it blocks as the filter is compressed or expanded."

The team used a special printer that scored paper to allow a sheet to be folded in the origami pattern. An inkjet-type printer was then used to apply lines of silver ink across those perforations, forming dipole elements that gave the object its RF filtering ability.

World's first AI-controlled wheelchair

Brazil's Hoobox Robotics has partnered with Intel on a prototype kit that exploits artificial intelligence (AI) to enable a disabled person to drive a motorised wheelchair using a collection of 10 facial expressions. Wheelie 7 learns these gestures automatically, and a caregiver can use an app to assign specific facial expressions to wheelchair movement or stopping commands.

Wheelie 7 allows people with movement-limiting conditions to get their autonomy back (Credit: Hoobox Robotics)

Wheelie 7 employs facial-recognition software, sensors, robotics and Intel's 3D RealSense depth camera mounted on the wheelchair. The system captures a 3D map of the user's face, and employs AI algorithms to process data in real time to control the wheelchair.

Hoobox's Paulo Pinheiro says that Wheelie 7 can operate in sunlight and dim-lighting conditions. It is compatible with 95 per cent of the commercially-available motorised wheelchairs.

Finney, a cyber-protected, blockchain-enabled smartphone

Sirin Labs has launched what it calls the world's first blockchain smartphone. Named after Bitcoin pioneer Hal Finney, the cyber-protected, blockchain-enabled smartphone comes with built-in cold storage wallet and unique Safe Screen feature for secure crypto transactions.

Finney is a blockchain smartphone, with a built-in cyber security suite, crypto wallets, DApps and token conversion service, all supported by SRN token—exclusive utility token to purchase Sirin Labs products and services (Credit: Sirin Labs)

Finney employs a multi-layered cyber security suite that includes a behaviour-based machine learning intrusion prevention system, for pro-active cyber protection in real time. It offers a decentralised app (DApp) store that enables users to use DApps, learn about new crypto projects, and earn new coins and tokens.

The smartphone is based on Sirin operating system (OS), which is the only OS secure enough for storing and using cryptocurrency in a mobile environment. It provides users with the proprietary token conversion service, which enables seamless and automatic exchange between supported tokens and coins, eliminating the need to visit external exchanges.